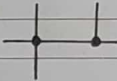
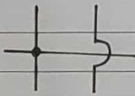
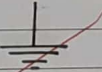


FAMILIARIZATION OF SYMBOLS

AIM

To familiarise and understand the general symbols in electrical and electronic circuit

Symbol	Component Name	Meaning
WIRE SYMBOLS		
	Electrical wire	conductor of electrical current
	connected wire	connected crossing
	not connected wire	Wires are not connected
GROUND SYMBOLS		
	earth ground	Used for zero potential (reference) and electrical shock potential
	chassis ground	connected to chassis of the circuit

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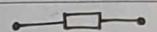
Digital / Common ground

RESISTOR SYMBOLS



Resistor (IEEE)

Resistor reduce the current flow



Resistor (IEC)



Potentiometer (IEEE)

Adjustable resistor have 3 terminals



Potentiometer (IEC)



Variable resistor / Rheostat (IEEE)

Adjustable resistor has 2 terminals

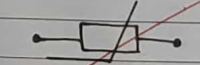


Variable resistor / Rheostat (IEC)



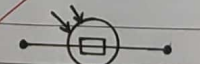
Trimmer resistor

Preset resistor



Thermistor

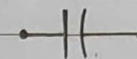
Thermal resistor - changes resistance when temperature changes.



Photoresistor / Light dependent resistor (LDR)

Teacher's Signature

CAPACITOR SYMBOLS

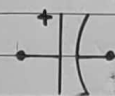


Capacitor

Capacitor is used to store electric charge. It acts as a short circuit with AC and an open circuit with DC.

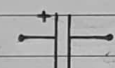


Capacitor

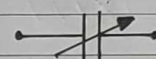


Polarised Capacitor

Electrolytic Capacitor



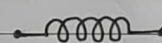
Polarised Capacitor



Variable Capacitor

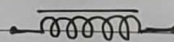
Adjustable Capacitor

Inductor / coil symbols



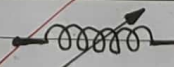
Inductor

Coil / Solenoid that generates magnetic field.



Iron core Inductor

Includes iron



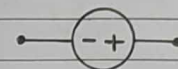
Variable Inductor

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POWER SUPPLY SYMBOLS



Voltage source

Generates constant voltage



Current source

Generates constant current



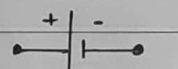
AC Voltage source

AC Voltage Source



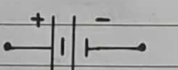
Generator

Electrical Voltage is generated by mechanical rotation of the generator



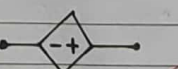
Battery Cell

Generates constant voltage



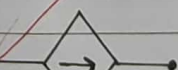
Battery

Generates constant voltage



Controlled Voltage source

Generates voltage as a function of voltage or current of other circuit elements



Controlled current source

Generates current as a function of voltage or current of other circuit elements

Teacher's Signature

METER SYMBOLS



Voltmeter

Measures voltage,
has high resistance connected
in parallel

Ammeter

Measures electric current
has zero resistance
connected serially

Ohmmeter

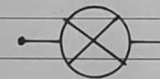
Measures resistance



Wattmeter

Measures electric power

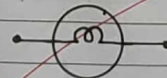
LAMP / LIGHT BULB SYMBOLS



light / lamp bulb



light / lamp bulb

generates light when
current flows through

light / lamp bulb

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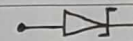
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DIODE / LED SYMBOLS



Diode

Diode allows current flow in one direction, left to right



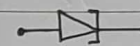
Zener Diode

allows current flow in one direction but can flow in reverse direction when above breakdown voltage

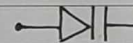


Schottky Diode

Schottky diode is a diode with a low voltage drop.



Tunnel Diode



Varactor / Varicap diode

Variable capacitor diode



Light emitting diode (LED)

LED emit light when current flow



Photodiode

Photo diode allows current flow when exposed to light.

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TRANSISTOR SYMBOLS

	NPN bipolar transistor	allows current flow when high potential of base (middle)
	PNP bipolar transistor	allows current flow when low potential at base (middle)
	JFET - N transistor	N channel field effect transistor
	JFET - P Transistor	P channel field effect transistor
	NMOS Transistor	N channel MOSFET Transistor
	PMOS Transistor	P channel MOSFET Transistor
	Darlington transistor	Mode of 2 bipolar transistor. has total gain of the product of each gain

MISC SYMBOLS

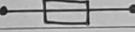
	Motor	electric motor
	Transformer	change AC voltage from high to low or low to high

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	Electric Bell	Rings when activated
	Buzzer	Produces buzzing sound
	Fuse	The fuse disconnects when current above threshold used to protect circuit from high current
	Fuse	

Supplementary Context, Symbol						Terms
SPST NO		SPST NC		SPDT		SPST - single pole single throw
Single Break	Double Break	Single Break	Double Break	Single Break	Double Break	SPDT - single pole double throw
						DPST - double pole single throw
DSPT NO		DSPT NC		DPDT		DPDT - double pole double throw
Single Break	Double Break	Single Break	Double Break	Single Break	Double Break	
						NO - Normally open
						NC - Normally closed

RESULT

Familiarized the general symbols in electrical and electronic circuits.

Teacher's Signature _____